Chapter 2: Review of Related Literature and Studies

This chapter reviews related literature and studies that ground the proposed benchmarking of dynamic multi-task balancing on Taglish disaster tweets. Following the required thematic organization, the review arranges prior work by computational sub-concept, algorithmic paradigm, and empirical bottleneck rather than by geographical origin.

Section 2.1 examines crisis informatics corpora and the evidence for joint multi-task modeling of Named Entity Recognition (NER), intent classification, and urgency classification. Section 2.1 also examines the computational and statistical standards that such modeling must satisfy. Section 2.2 analyzes multi-task optimization (MTO) dynamics, covering dynamic loss-weighting methods, gradient-surgery and projection techniques, and the empirical evidence questioning whether specialized balancing algorithms outperform static scalarization. Section 2.3 reviews code-switched text processing, subword tokenization disparities, and Philippine language resources, including local disaster response studies. Section 2.4 consolidates key studies into an Algorithmic Literature Taxonomy Matrix positioned against the proposed research. Section 2.5 presents a three-step synthesis that derives the research gap and articulates how the proposed study bridges that gap.

The reviewed corpus draws on peer-reviewed sources indexed in the ACM Digital Library, IEEE Xplore, SpringerLink, ScienceDirect, and the Association for Computational Linguistics (ACL) Anthology. Sources were identified through database searches on multi-task optimization, crisis informatics, and code-switching conducted as of 2026. Foundational multi-task optimization studies from 1997 to 2018 are retained where they define the algorithms under benchmark. The majority of the crisis informatics and code-switching sources that establish the evaluation domain date from the last five years.

2.1 Joint Multi-Task Modeling and Computational Efficiency in Crisis Informatics

2.1.1 Crisis Informatics Corpora and Benchmark Protocols

Crisis informatics research established that human-annotated social media corpora enable supervised processing of disaster messages at operational scales. Imran et al. (2016) collected over 52 million crisis-related tweets across 19 disaster events and released approximately 50,000 human-annotated messages. These messages spanned nine United Nations Office for the Coordination of Humanitarian Affairs (UN OCHA) humanitarian categories. Across Naive Bayes, Random Forest, and Support Vector Machine classifiers, the corpora achieved Area Under the ROC Curve (AUC) scores of at least 0.80 for most humanitarian categories. The rare "missing, trapped, or found people" category suffered measurably lower performance because of severe class imbalance. The authors additionally documented five categories of out-of-vocabulary noise in crisis microblogs, including misspellings, abbreviations, and phonetic substitutions, confirming that informal disaster text requires specialized lexical handling.

Subsequent benchmarks standardized label taxonomies, deduplication, and evaluation splits. Alam, Sajjad, et al. (2021) consolidated eight crisis tweet corpora into CrisisBench and demonstrated that duplicate and near-duplicate content is pervasive in uncleaned crisis datasets. Failing to remove such content therefore produces misleading classification performance and risks data leakage between training and evaluation splits. On the consolidated humanitarian task, RoBERTa achieved a weighted F1-score of 0.872 compared with 0.829 for the best classical baseline. Preliminary multilingual experiments found that XLM-RoBERTa matched monolingual RoBERTa on the consolidated multilingual set, while systematic multilingual evaluation was left as future work. Alam, Qazi, et al. (2021) released HumAID, a 77,196-tweet corpus covering 19 disaster events between 2016 and 2019. The corpus was filtered from 24.8 million raw tweets through geographic, relevance, and near-duplicate pruning. Crowdsourced annotation reached an average Fleiss' kappa of 0.55. The HumAID benchmark showed that monolingual RoBERTa attained the highest average weighted F1-score of 0.781, followed by multilingual XLM-RoBERTa at 0.777, whereas classical classifiers remained below 0.72. Notably, DistilBERT matched full BERT performance (0.769 versus 0.768 average weighted F1) with substantially fewer parameters, a result the authors identified as directly relevant to latency-sensitive operational deployment.

Recent resources extend document-level corpora toward token-level annotations and multi-task formulations. C. Wang et al. (2021) formulated disaster triage as multi-task learning on the TREC Incident Streams corpus (40,459 tweets across 42 crisis events). The model jointly predicts 25 information types and a continuous priority score. The shared BERT encoder outperformed single-task learning on ranking quality (NDCG 0.5101 versus 0.4393) and alerting worth (-0.2689 versus -0.4057 for high-priority alerts). The encoder ablation showed that parameter-efficient variants (DistilBERT at 66 million parameters and

ALBERT at 11 million parameters) remained competitive with full 110-million-parameter encoders. However, the study did not evaluate computational efficiency or inference latency. Its fixed equal weighting is precisely the untested condition that Section 2.2 shows can suppress one task family in favor of another. Token-level NER aggregates sequence-labeling gradients across every subword token, whereas sequence-level classification contributes a single pooled signal. Gradient-magnitude disparity theory predicts that this asymmetry will degrade the weaker task's gradients (Chen et al., 2018). No corpus in this lineage pairs token-level entity spans with concurrent sequence-level triage labels under cross-event evaluation, a void the proposed study fills directly.

2.1.2 Joint Versus Single-Task Modeling of Token-Level and Sequence-Level Tasks

Foundational work in multi-task learning (MTL) established the representational and statistical rationale for joint training. Caruana (1997) formalized hard parameter sharing, in which related tasks train in parallel over a common hidden representation. Caruana also demonstrated on road-following and medical risk prediction domains that the inductive transfer improves generalization on the primary task. Baxter (2000) proved statistically that learning a shared representation across K related tasks reduces the per-task sample complexity by a factor proportional to $1/K$. This reduction holds provided that tasks originate from a common environment distribution. Ruder (2017) systematized these results for deep networks, delineating hard from soft parameter sharing and identifying five inductive transfer mechanisms, including implicit data augmentation, eavesdropping, and representation bias.

The survey also flagged the unresolved problem of predicting task compatibility and mitigating conflicting gradients. Sections 2.2 and 2.3 examine these two failure modes empirically. Zhang & Yang (2021) confirmed through a broad survey that deep multi-task models now dominate computer vision and natural language processing applications. However, negative transfer from unrelated or outlier tasks remains a persistent theoretical weakness lacking principled mitigation (Ruder, 2017).

Empirical studies of joint sequence labeling and sequence classification provide direct precedent for the proposed NER, intent, and urgency task bundle. A. Wang et al. (2019) introduced the GLUE multi-task benchmark and reported that multi-task training with attention-based BiLSTM encoders reached a 70.0 macro-average versus 67.4 for single-task counterparts. However, diagnostic probing revealed destructive interference when tasks were trained jointly under uniform sampling. X. Liu et al. (2019) demonstrated with MT-DNN that multi-task fine-tuning of BERT raised the GLUE benchmark score to 82.7 (versus 80.5 for BERT-large). Fine-tuning also enabled domain adaptation with 82.1 percent accuracy on SNLI using only 0.1 percent of in-domain samples. The authors attributed the gains to MTL-induced regularization. In task-oriented dialogue, Weld et al. (2022) surveyed joint intent detection and slot filling. Joint models consistently outperformed independent pipelines by capturing mutual conditional distributions and eliminating cascading error propagation. Slot F1 on the ATIS benchmark improved steadily across successive systems, and intent accuracy exceeded 99 percent. Weld et al. (2022) cautioned, however, that long-solved benchmarks increasingly measure dataset difficulty rather than language understanding. The survey identified standardized training budgets and out-of-domain generalization as open problems for the field. These findings jointly support the premise that pairing token-level NER with sequence-level intent and urgency heads offers strong mutual reinforcement. The benefits of pairing nevertheless depend on task relatedness, dataset scale, and optimization control.

2.1.3 Complexity Analysis, Statistical Rigor, and Reproducibility Infrastructure

The computational case for hard parameter sharing follows from asymptotic complexity analysis. Deploying K independent single-task models requires K forward passes per input and $O(K \cdot |\theta_{\text{backbone}}|)$ parameter storage. A shared encoder computes all task features in one forward pass, with task heads contributing less than one percent of total parameters. This architecture yields effective $\mathcal{O}(1)$ time and memory relative to task count K (Caruana, 1997; Ruder, 2017; Zhang & Yang, 2021). Lin & Zhang (2023) operationalized this architecture family in the LibMTL PyTorch library, which unifies 13 weighting strategies and 8 sharing architectures. Their NYUv2 benchmarks showed that exact gradient-balancing methods demand per-task gradients over all shared parameters and require representation-level approximations when parameter counts grow. The library initially lacked natural language processing backbones.

Two methodological studies define the statistical standards that the proposed benchmark adopts. Reimers & Gurevych (2017) retrained published sequence-tagging systems across dozens of random seeds and showed that single reported scores are statistically misleading. Score variation across random seeds reached 1.01 F1 on CoNLL-2003 NER and 8.23 F1 on ACE 2005. Development-set selection correlated only weakly with test performance

(Spearman $\rho = 0.229$). Demšar (2006) demonstrated that classifier comparisons across multiple datasets violate the normality and sphericity assumptions of parametric tests. The study recommended the Friedman test with the Iman-Davenport correction followed by post-hoc Nemenyi procedures as the default non-parametric protocol. These two studies jointly motivate the proposed design decisions of multi-seed evaluation under Leave-One-Event-Out (LOEO) cross-validation and two-tier non-parametric significance testing.

2.2 Multi-Task Optimization Dynamics: Balancing Conflicting Task Gradients

2.2.1 Gradient Surgery and Projection Techniques

Sener & Koltun (2018) reframed multi-task learning as multi-objective optimization and adapted the Multiple Gradient Descent Algorithm (MGDA) with a Frank-Wolfe line search to locate Pareto-stationary updates. Exact MGDA scales linearly with task count because the algorithm requires separate backward passes. The authors therefore proposed MGDA-UB, which solves the min-norm problem over representation gradients in a single backward pass. MGDA-UB accelerated training by a factor of 25 on the 40-task CelebA benchmark. Yu et al. (2020) identified the "tragic triad" of high curvature, disparate gradient magnitudes, and conflicting directions, and proposed Projecting Conflicting Gradients (PCGrad). PCGrad projects each task gradient onto the normal plane of any conflicting gradient with negative cosine similarity. PCGrad raised Multi-task CIFAR-100 accuracy to 71.0 percent (versus 67.7 percent for independent training). It also solved all ten Meta-World MT10 manipulation tasks with multi-task Soft Actor-Critic where the vanilla joint agent failed half of the tasks. Ablations showed that altering both gradient directions and magnitudes is necessary. The authors acknowledged two structural costs. First, convergence can stall when cosine similarity approaches -1. Second, computing per-task gradients multiplies backward passes linearly with task count.

B. Liu et al. (2021) strengthened the convergence guarantees with Conflict-Averse Gradient Descent (CAGrad). CAGrad maximizes the worst-case local task improvement within a trust region around the average gradient and provably converges to the average-loss optimum when the hyperparameter satisfies $c < 1$. CAGrad halved the average performance drop on CityScapes (11.64 percent versus 22.60 percent for shared MTAN under equal weighting) and matched or surpassed PCGrad across NYUv2 and Meta-World. A task-sampled variant delivered 2x to 5x speedups. L. Liu et al. (2021) approached fairness directly with Impartial MTL (IMTL). IMTL-G computes closed-form scaling factors so that the aggregated gradient makes equal projections onto every raw task gradient. IMTL-L rebalances loss scales through learnable parameters without distributional assumptions. Both variants remain invariant to manual loss rescaling and are computable through a single backward pass over shared representations. Navon et al. (2022) grounded gradient aggregation in cooperative bargaining theory, deriving Nash-MTL as the unique scale-invariant update satisfying Pareto optimality, symmetry, and independence of irrelevant alternatives. Nash-MTL achieved the best overall trade-off on QM9 molecular regression (mean rank 2.5 versus 6.8 for linear scalarization, whose Δm reached 177.6 percent). On NYUv2, it reached a Δm of -4.04 percent versus $+0.20$ percent for CAGrad. However, each update requires solving a K-dimensional convex problem. The authors mitigated this cost by refreshing the bargaining solution only every 50 to 100 steps at a modest performance cost.

Recent methods continue to rederive balancing from structural analyses of gradients and kernels. Examples include singular-value alignment through orthogonal Procrustes transformations, projected-norm imbalance constraints, Neural Tangent Kernel eigenvalue weighting, and architectural layer duplication (Qin et al., 2025; Senushkin et al., 2023; Shi et al., 2023; Zhou et al., 2025). However, none of these variants has been evaluated on language tasks, let alone code-switched text. All of them inherit the per-task backward-pass economics described above. Read together, the gradient-surgery family exhibits a common structural cost. PCGrad requires per-task backward passes, CAGrad adds a trust-region subproblem, and Nash-MTL solves a K-dimensional convex program per update. Each surgical intervention therefore adds training-time overhead that scales with task count. This cost is precisely what the empirical skepticism of Section 2.2.3 interrogates.

2.2.2 Dynamic Loss-Weighting Methods

An alternative algorithmic family adapts scalar loss weights without modifying gradient directions. Kendall et al. (2018) derived homoscedastic uncertainty weighting from a probabilistic likelihood in which each task receives a learnable variance parameter, producing the loss term $\frac{1}{2\sigma_i^2}\mathcal{L}_i + \log \sigma_i$. On CityScapes, uniform weighting degraded semantic segmentation to 50.1 percent IoU, below the 59.4 percent single-task level. Learned uncertainty weighting recovered 63.4 percent IoU with a final weight ratio of 43:1:0.16 across segmentation, depth, and instance tasks. It outperformed grid-searched static weights (62.8 percent

IoU) because the weights adapt during training. Chen et al. (2018) proposed GradNorm, which adjusts loss weights so that gradient norms at the final shared layer track relative inverse training rates. A single GradNorm run matched or surpassed the results of a 100-run weight grid search on NYUv2 (depth RMS error of 0.925 m at $\alpha = 1.5$). This run required approximately 5 percent computational overhead. S. Liu et al. (2019) introduced Dynamic Weight Average (DWA) within the Multi-Task Attention Network (MTAN). DWA balances tasks from prior-epoch loss descent rates alone and requires no additional backward passes. This design made training robust to the choice of weighting scheme because attention-based feature selection implicitly balances tasks. B. Liu et al. (2023) subsequently proposed FAMO, which equalizes task log-loss improvement rates through a first-order update on weighting logits. FAMO amortizes all gradient computations into a single backward pass, achieving $\mathcal{O}(1)$ time and space per iteration. FAMO outperformed linear scalarization on NYUv2 (Δm of -4.10 percent versus $+5.59$ percent) and CelebA (40 tasks). It trained 25 times faster than Nash-MTL on a two-task synthetic benchmark. The authors acknowledged reliance on temporal loss differences as directional-derivative estimates and the need to tune the regularization decay hyperparameter γ.

2.2.3 Empirical Skepticism: Scalarization Baselines and Optimizer Confounds

A converging body of evidence questions whether specialized MTO algorithms justify their costs when baselines are tuned properly. Kurin et al. (2022) benchmarked unitary (equal-weight) scalarization against MGDA, GradNorm, PCGrad, CAGrad, and IMTL across vision, reinforcement learning, and machine translation benchmarks under standardized hyperparameter budgets. Well-tuned equal weighting with Adam matched or exceeded every specialized method. The specialized algorithms ran up to 35 times slower because of their multiple backward passes. The authors interpreted many balancing schemes as implicit regularizers rather than superior optimizers. Xin et al. (2022) reached congruent conclusions on large-scale multilingual machine translation and vision. Dynamic task weights assigned by MTO algorithms remained nearly constant during training. Per-task gradient computation reduced training throughput from 11.52 to 4.81 steps per second. Reallocating that budget to model depth strictly improved the Pareto frontier. In addition, learning-rate underspecification produced six to seven times more score variance than seed variation, creating illusions of algorithmic superiority in undertuned comparisons.

Elich et al. (2024) identified the standard optimizer itself as the decisive confound. Across CityScapes, NYUv2, and CelebA, Adam outperformed SGD with momentum for every architecture and weighting scheme. The authors proved that Adam is partially loss-scale invariant for task-specific heads because loss scaling factors cancel between the first- and second-moment updates. Optimal uncertainty weighting is fully scale invariant. Under Adam, plain Equal Weighting (EW) achieved Pareto-optimal performance in three of four benchmark configurations and was not dominated by any specialized MTO method trained with SGD. Most consequentially for diagnostic methodology, Elich et al. (2024) measured gradient cosine similarities between samples within a single task. Intra-task, inter-sample conflicts occurred as frequently as cross-task conflicts. Gradient magnitude disparities, not angular misalignment, constitute the primary optimization challenge under adaptive optimization. Task-selection studies reinforce this caution. Standley et al. (2020) showed that joint-training task affinity is uncorrelated with transfer affinity (Pearson $\rho = -0.12$). The study also showed that all-in-one networks are often suboptimal under inference time budgets. Collectively, these results indicate that claims of balancing gains must be established under optimizer-controlled, multi-seed, budget-matched conditions on the target domain. The proposed Taglish triage benchmark adopts these conditions by design. It extends the audit protocol of Elich et al. (2024) and Kurin et al. (2022) to code-switched crisis text.

2.3 Code-Switched Text Processing and Low-Resource Philippine NLP

2.3.1 Computational Challenges of Code-Switched Language

Systematic surveys converge on structural obstacles that confront code-switched natural language processing. Doğruöz et al. (2021) argued that computational work largely ignores grammatical and sociolinguistic theories of switching. They argued that multilingual pretrained models underperform on code-switched inputs because pretraining corpora lack naturalistic mixed language. They also noted that benchmarks cover predominantly Spanish-English and Hindi-English pairs. Winata et al. (2023) reviewed more than 400 publications across three decades. The review documented severe resource concentration around English-paired bilingual data, dominance of language identification and sentiment tasks, and dataset attrition. Over 60 percent of surveyed corpora are no longer publicly accessible. Sheth, Sinha, et al. (2025) synthesized 327 studies spanning the large language model era and reported three findings. Large language models frequently underperform smaller fine-tuned task-specific models on code-switched inputs. In addition, over 72 percent of speech and 92 percent of social media code-switching datasets involve English-centric pairs. Standard generation metrics also correlate poorly with human judgments on mixed-language output. Southeast Asian and Austronesian contact varieties, including Tagalog-English code-switching (Taglish), remain critically underrepresented in all three surveys.

Subword tokenization quantifies a concrete computational penalty for underrepresented contact varieties. Ahia et al. (2023) measured token fertility across commercial tokenizers and showed that identical semantic content in low-resource languages requires 2x to 7x more subword tokens than English. This inflation raises per-token API costs and inference latency. Petrov et al. (2023) extended the analysis across open multilingual tokenizers and documented sequence-length inflation of up to 15x for non-Latin scripts, which prematurely exhausts context windows and increases latency. For a Taglish triage pipeline, these findings imply that subword over-fragmentation dilutes token-level embeddings for NER, where span labels must survive fragmentation across more subword pieces. Over-fragmentation also increases optimization variance precisely where supervision is densest. These findings motivate multilingual encoders whose vocabularies cover Tagalog-English mixing.

2.3.2 Multi-Task Learning on Code-Switched Text

Small but growing literature tests multi-task architectures on code-switched text, and the findings remain mixed. Khanuja et al. (2020) released GLUECoS, a six-task benchmark for English-Hindi and English-Spanish code-switching. The study showed that mBERT fine-tuned on a synthetic-then-real code-switched curriculum raised NER F1 to 78.21 (versus 74.96 for vanilla mBERT), while natural language inference remained near chance. The authors explicitly identified a unified multi-task learning setup for cross-task transfer as unexecuted future work. Sheth, Beniwal, et al. (2025) constructed COMI-LINGUA, 125,615 expert-annotated Hindi-English instances with Fleiss' kappa of at least 0.817 across five tasks. The study showed that parameter-efficient fine-tuning of 7-8B models reached up to 95.25 NER F1, exceeding zero-shot prompting of commercial large language models. Most directly relevant to the proposed research, Adouane & Bernardy (2020) conducted controlled pairwise MTL experiments on low-resource, noisy, code-switched Algerian user-generated text. The tasks spanned code-switch detection, NER, spelling normalization, and sentiment analysis. The results demonstrated that multi-task benefits are conditional rather than universal. Pairing NER with sentiment analysis induced negative transfer that collapsed NER macro F1 from 49.68 to 34.60. In contrast, pairing NER with code-switch detection produced mutual gains that raised detection macro F1 from 64.54 to 71.29. The authors further documented that joint-training outcomes were unstable across task pairings and dataset sizes. The mechanism most conspicuously absent from their experimental controls was an explicit loss-balancing procedure. This evidence establishes both the promise and the risk of the proposed three-task Taglish bundle. It supplies the strongest empirical motivation for benchmarking explicit MTO algorithms rather than assuming automatic joint-training benefits. It additionally motivates the proposed study's measurement of per-task-pair gradient cosine similarity. The Algerian results imply that the NER-intent and NER-urgency pairs may behave differently from the intent-urgency pair. A more recent preprint study applied hard parameter sharing with homoscedastic uncertainty weighting to code-switched NER and POS tagging across three language pairs from the LinCE benchmark. It reported NER F1 of 71.4 to 82.9 under 10-fold cross-validation (S R et al., 2023). Although preliminary and not yet peer-reviewed, the study corroborates uncertainty weighting as the natural first baseline for code-switched multi-task objectives.

2.3.3 Philippine Language Resources and Local Disaster Response Studies

Philippine NLP has matured from vocabulary-scale models to benchmark infrastructure, but along a trajectory that bypasses disaster-domain multi-tasking. Cruz & Cheng (2022) pretrained RoBERTa Base (110M) and Large (330M) models on the TLUnified corpus. The new models improved test accuracy over prior Filipino BERT baselines by an average of 4.47 points (72.67 percent on election hate speech detection versus 68.60 percent). Scaling from Base to Large yielded only marginal gains (+0.09 to +0.90 points) because of pretraining corpus size. The evaluation was delimited to classification without sequence labeling. Miranda (2023) addressed sequence labeling with TLUNIFIED-NER, a 208,247-token news corpus annotated for Person, Organization, and Location spans. The corpus has inter-annotator Cohen's kappa of 0.81 to 0.83 and reference F1 of 0.89. Fine-tuned supervised models consistently outperformed zero-shot prompting of commercial and open large language models. The corpus covers formal news text rather than informal code-switched social media. Montalan et al. (2025) introduced Batayan, an eight-task Filipino and Taglish benchmark of 3,800 natively validated test instances. The benchmark showed that Southeast Asian instruction-tuned models such as Gemma-SEA-LION-v3-9B reached an average macro F1 of 79.23 on natural language understanding tasks. Its evaluation is task-isolated and excludes the disaster domain. Herrera et al. (2022) contributed TweetTaglish, 21,150 code-switched tweets with token- and tweet-level language distribution annotations. The average composition comprises 55.2 percent Tagalog, 19.4 percent English, and 24.5 percent other tokens, validated at instance-level kappa of 0.70. Proof-of-concept regressors reached R^2 of 0.909 for English proportion prediction. The resource contains no triage annotations. On the encoder side, Warner et al. (2025) demonstrated with ModernBERT that hardware-aware depth, full-sequence unpadding, and alternating local-global attention deliver 2.65x to 3.0x throughput gains at long contexts. These findings inform efficient encoder selection for the SEA-LION ModernBERT candidate backbone. The authors delimited the model family to English text and code.

Local disaster-focused studies address single tasks on single events with non-Transformer models. Imperial et al. (2019) classified sentiment in 3,900 annotated Typhoon Yolanda (Haiyan) tweets with standard and bidirectional recurrent networks. The models reached 81.79 percent accuracy for three-class and 87.69 percent accuracy for binary classification. False negatives accounted for 12 to 14 percent of errors. Ermino et al. (2022) trained a lightweight feedforward artificial neural network on 4,020 Typhoon Ulysses (Vamco) tweets for binary urgency evaluation. The model reported 99.17 percent validation accuracy and an F1 of 0.9882 on a 100-tweet confusion-matrix set. The authors acknowledged a fixed 8,000-word vocabulary and a single-event training distribution as limitations. On the urgency task specifically, Kejriwal & Zhou (2019) built low-supervision urgency classifiers for short crisis messages by ensembling keyword features with locally trained and pre-trained embeddings. The study demonstrated a transfer variant trained on pooled source events and evaluated on held-out events. This protocol anticipates, at smaller scale, the cross-event evaluation design of the proposed study. At the cross-lingual frontier, Sarioglu Kayi et al. (2020) transferred urgency classification from English to Sinhala and Odia. The model reached 75.7 macro F1 on English with RoBERTa-large but only 63.5 and 62.6 zero-shot macro F1 on the target languages. The annotators conflated general status updates with actionable distress calls. Ray Ray Chowdhury et al. (2020) aggregated more than 130,000 crisis tweets and applied Manifold Mixup to multilingual BERT. The approach improved out-of-event generalization across four held-out disaster test sets, including 79.36 macro F1 on the Philippines Flood test event. Per-class analysis showed Caution-and-Advice as a comparatively weak class at 70.3 F1. Across this body of work, no study combines token-level entity extraction, intent categorization, and urgency prioritization on code-switched Philippine disaster text under cross-event evaluation.

2.4 Algorithmic Literature Taxonomy Matrix

Table 1 positions the principal related studies against the proposed research across algorithm, dataset or testbed, evaluated metrics, and identified limitations. The matrix organizes studies into four thematic clusters. Crisis corpora and multi-task triage occupy rows 1 to 4. MTO algorithms occupy rows 5 to 14. Code-switched and Philippine NLP occupy rows 15 to 17. The proposed study occupies row 18. Each identified limitation states the concrete absence that the proposed study supplies.

Table 1

Algorithmic Literature Taxonomy Matrix

No.	Author (Year)	Algorithm / Model Used	Dataset / Testbed	Evaluated Metrics	Identified Limitation / Gap
1	Imran et al. (2016)	NB, RF, SVM over domain CBOW embeddings	19 disasters; 52M tweets, ~50K annotated	AUC, 10-fold cross-validation	English-only; class imbalance on rare categories; no token-level spans; no joint tasks
2	Alam, Sajjad, et al. (2021)	CNN, fastText, BERT, RoBERTa fine-tuning	CrisisBench consolidated crisis tweets	Weighted precision, recall, F1, accuracy	Single-label scheme; no entity or urgency annotations; multilingual evaluation preliminary
3	Alam, Qazi, et al. (2021)	fastText, BERT, DistilBERT, RoBERTa, XLM-R	HumAID: 77,196 tweets, 19 events	Weighted precision, recall, F1; inter-annotator agreement	English-only; mutually exclusive labels; kappa 0.55; no latency or memory reporting
4	C. Wang et al. (2021)	Shared BERT MTL (25-class info types + priority regression)	TREC-IS: 40,459 tweets, 42 events	NDCG, alert worth, F1, accuracy, harmonic mean	Equal weighting only; no gradient diagnostics; efficiency, latency, and throughput unmeasured
5	Kendall et al. (2018)	Homoscedastic Uncertainty Weighting	CityScapes (vision)	Segmentation IoU, instance AP, centroid pixel error, depth RMSE, weight trajectories	Vision-only; homoscedastic noise assumption; no text or NLP evaluation
6	Chen et al. (2018)	GradNorm (gradient-norm targets via training rates)	NYUv2, synthetic regression (vision)	Task-normalized loss, depth RMSE, segmentation IoU, surface normal error, keypoint error	Balances only the final shared layer; ~5% overhead; no language tasks
7	Yu et al. (2020)	PCGrad (orthogonal projection of conflicting gradients)	CIFAR-100, NYUv2, Meta-World MT10	Accuracy, classification error, segmentation mIoU, depth error, task success rate	K backward passes; order-dependent projections; stall risk at cosine -1; no NLP
8	B. Liu et al. (2021)	CAGrad (trust-region worst-case improvement)	NYUv2, CityScapes, MT10 (vision)	Training loss, per-task performance, mIoU, depth error, success rate, training time	Trust-region hyperparameter c requires per-dataset tuning; no NLP evaluation
9	L. Liu et al. (2021)	IMTL-G / IMTL-L (equal-angle projection + loss balance)	NYUv2, CityScapes, CelebA (vision)	mIoU, depth error, normals error, accuracy	Assumes equal task priority; vision-only; no gradient-conflict logging
10	Navon et al. (2022)	Nash-MTL (Nash bargaining over task gradients)	NYUv2, CityScapes, QM9, MT10	Mean rank, Δm, per-task metrics, runtime	Per-step K-dimensional convex solver; cost mitigated by infrequent updates; no code-switched text
11	B. Liu et al. (2023)	FAMO (log-loss improvement equalization, $\mathcal{O}(1)$)	NYUv2, CityScapes, QM9, CelebA, MT10	Δm, mean rank, runtime vs scalarization	Decay hyperparameter γ requires tuning; temporal-difference approximation; no NLP
12	Kurin et al. (2022)	Unitary scalarization vs MGDA, GradNorm, PCGrad, CAGrad, IMTL	NYUv2, CityScapes, MT10	Per-task metrics, wall-clock runtime, GPU memory	Specialized methods up to 35x slower; scope limited to moderately related tasks; no code-switched NLP
13	Xin et al. (2022)	Empirical audit of MTO vs static scalarization	WMT translation, CityScapes, CelebA	BLEU, mIoU, error rates, steps/sec (11.52 to 4.81)	Dynamic weights nearly static in practice; Pareto grid search prohibitive; no low-resource text
14	Elich et al. (2024)	Equal Weighting + Adam; loss-scale invariance proofs	CityScapes, NYUv2, CelebA	Δm, mIoU, depth errors, gradient cosine/magnitude statistics	Vision-only; gradient statistics subsampled; language tasks untested
15	Khanuja et al. (2020)	mBERT + code-switched curriculum (GLUECoS)	English-Hindi, English-Spanish benchmarks	Token F1, accuracy, QA F1, NLI accuracy	Single-task evaluation only; MTL setup left as future work; no triage tasks

No.	Author (Year)	Algorithm / Model Used	Dataset / Testbed	Evaluated Metrics	Identified Limitation / Gap
16	Adouane & Bernardy (2020)	Shared-CNN MTL across CSD, NER, SPELL, SA	Algerian Arabic user-generated text	Accuracy, macro/micro F1	No principled loss balancing; task pairing decisive (NER+SA macro F1 fell 49.68 to 34.60)
17	Herrera et al. (2022)	Dictionary language ID + regression benchmark (MLP, SVR)	TweetTaglish: 21,150 Taglish tweets	R^2, RMSE, inter-annotator kappa	Language-distribution labels only; word-level kappa 0.15 from cognate noise; no triage tasks
18	Proposed study	Dynamic loss weighting (Uncertainty Weighting, GradNorm, FAMO) and gradient surgery (PCGrad, CAGrad, IMTL-G, Nash-MTL) versus STL, Uniform Equal Weighting, and tuned Static Linear Scalarization on a shared multilingual Transformer encoder	Multi-Task Corpus of Taglish Disaster Tweets (4 typhoon events; 4-fold LOEO; 3 seeds; RTX 4060 8GB workstation testbed)	Macro F1 (intent, urgency), span F1 (NER), Δm, t_{exec} (B=1), throughput (B=32), peak VRAM/RAM, training runtime, gradient norms and pairwise cosine similarities	Addresses the literature gap directly: first MTO benchmark on code-switched Taglish disaster triage with concurrent token- and sequence-level annotations, gradient-conflict diagnostics against an intra-task half-batch baseline, cross-event validation, and latency and memory profiling under AdamW optimization

Note. MTL = multi-task learning. MTO = multi-task optimization. NER = named entity recognition. STL = single-task learning. CSD = code-switch detection. SPELL = spelling normalization. SA = sentiment analysis. NLI = natural language inference. QA = question answering. NDCG = normalized discounted cumulative gain. Δm = relative multi-task performance change.

2.5 Synthesis of Related Literature and Studies

Three established trends emerge across the reviewed literature. First, crisis informatics has converged on Transformer encoders as the state-of-the-art backbone for humanitarian message processing. Monolingual RoBERTa-class models lead English benchmarks (weighted F1 of 0.872 on CrisisBench and 0.781 on HumAID). DistilBERT-class variants demonstrate comparable accuracy at lower parameter counts for deployment-oriented settings (Alam, Sajjad, et al., 2021; Alam, Qazi, et al., 2021). Second, joint multi-task modeling of token-level and sequence-level tasks is an accepted methodology that outperforms independent pipelines when tasks are compatible. Evidence includes intent detection and slot filling (Weld et al., 2022) and disaster categorization with shared encoders (C. Wang et al., 2021). Hard parameter sharing provides effective $\mathcal{O}(1)$ inference cost relative to task count (Caruana, 1997; Ruder, 2017; Zhang & Yang, 2021). Third, the MTO literature has produced two mature algorithmic families. The first is dynamic loss weighting (Chen et al., 2018; Kendall et al., 2018; B. Liu et al., 2023). The second is gradient surgery (B. Liu et al., 2021; L. Liu et al., 2021; Navon et al., 2022; Qin et al., 2025; Senushkin et al., 2023; Yu et al., 2020; Zhou et al., 2025). Both families have been evaluated with increasing methodological rigor under multi-seed, non-parametric protocols (Demšar, 2006; Reimers & Gurevych, 2017). For code-switched and low-resource text, the field additionally agrees on prerequisites for usable systems (Ahia et al., 2023; Doğruöz et al., 2021; Khanuja et al., 2020; Petrov et al., 2023). The prerequisites are multilingual encoders, careful annotation validation, and tokenization-aware design.

Four unresolved bottlenecks persist at the intersection of these trends. First, the central empirical dispute in MTO remains unresolved. Tuned static scalarization with adaptive optimizers matches or exceeds specialized balancing algorithms on vision and translation benchmarks (Elich et al., 2024; Kurin et al., 2022; Xin et al., 2022). However, every audit study operated on well-resourced, monolingual, or vision data. The question of whether dynamic balancing earns its runtime overhead on low-resource, code-switched crisis text therefore remains unanswered. Second, the disaster MTL studies closest to the proposed problem stop short of optimization control. C. Wang et al. (2021) applied fixed weighting without gradient diagnostics or efficiency measurements. The study did not report inference latency, GPU memory, or cross-event generalization. The token-versus-sequence optimization behavior of disaster MTL therefore remains undiagnosed. Third, no existing corpus combines token-level entity spans, intent categories, and urgency tiers on code-switched Taglish disaster text. Philippine resources are task-isolated, formal-domain, or language-ID-oriented (Cruz & Cheng, 2022; Herrera et al., 2022; Miranda, 2023; Montalan et al., 2025). Local disaster studies address single tasks on single events with non-Transformer models (Ermino et al., 2022; Imperial et al., 2019). In addition, prior code-switched MTL evidence warns that unconditional joint training can collapse minority task performance (Adouane & Bernardy, 2020). Fourth, diagnostic confounding remains unaddressed in applied settings. Elich et al. (2024) showed that intra-task, inter-sample gradient conflicts rival cross-task conflicts. However, the decoupling procedure exists only in computer vision. No crisis informatics study separates genuine cross-task interference from mini-batch variance when justifying balancing overhead.

The proposed study bridges these gaps through four concrete contributions mapped directly onto the bottlenecks. To resolve the unresolved MTO dispute on the target domain, the study benchmarks seven candidate algorithms against isolated Single-Task Learning, Uniform Equal Weighting, and tuned Static Linear Scalarization. The seven candidates span both algorithmic families, dynamic loss weighting (Uncertainty Weighting, GradNorm, FAMO) and gradient surgery (PCGrad, CAGrad, IMTL-G, Nash-MTL). The evaluation applies an equal hyperparameter search budget, AdamW optimization (Loshchilov & Hutter, 2019), and a 4-fold Leave-One-Event-Out protocol across three seeds. This design directly extends the audit conditions of Elich et al. (2024) and Kurin et al. (2022) to code-switched crisis text. To supply the missing optimization evidence in disaster MTL, the framework integrates routines for logging gradients in PyTorch. These routines record per-task gradient norms, norm ratios, and pairwise cosine similarities on shared encoder layers. The framework pairs them with an intra-task baseline derived from half-batch gradients. This baseline operationalizes the decoupling principle of Elich et al. (2024) for crisis triage and enables Spearman correlation analysis between transfer gains Δm and measured conflict dynamics. To close the dataset void, the study constructs the Multi-Task Corpus of Taglish Disaster Tweets. The corpus provides concurrent, expert-adjudicated annotations for token-level entity spans, intent categories, and urgency tiers across four historical Philippine typhoons. This corpus is the first to pair the sequence-labeling depth of TLUNIFIED-NER (Miranda, 2023) with the code-switched linguistic coverage of TweetTaglish (Herrera et al., 2022). Finally, to ground deployment relevance, the benchmark records single-instance latency, batched throughput, and peak GPU and host memory under FP16 execution on an 8GB workstation. These measurements convert the deferred efficiency questions of C. Wang et al. (2021) into measured quantities. The study also integrates the resulting model into a prototype web application for disaster triage as the empirical demonstration vehicle required by Design Science Research evaluation. Through these four bridges, the study converts an unresolved optimization question, an undiagnosed gradient asymmetry, a missing multilingual corpus, and an unmeasured deployment cost into a single reproducible empirical benchmark. This benchmark targets multi-task crisis triage on Taglish disaster tweets.

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